**Bank Churn Analysis – DAX & Advance Power BI topics covered**

**Objective of the Project:**

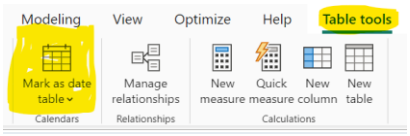
Analyse the data and bring out few insights on the customer Churn. It is advantageous for banks to know what leads a client towards the decision to leave the Bank.

**Calculations, measures (DAX):**

- **Additional new Tables and Columns:**
  Data has Dates – when data has dates, we must create date table & add required column like year, month, day etc.
  Always make master date table in that case to perform any time intelligent analysis.
  1. **Add new table (date table)**

```
Date Master Table = CALENDAR (FIRSTDATE (Bank Churn [Bank DOJ]), LASTDATE (Bank Churn [Bank DOJ]))
```

**Note:** now mark it as a date table




**Add year column,** `Year = YEAR ('Date Master Table'[Date])`

**Add Month column,** `Month = MONTH ('Date Master Table'[Date])`

**Add month name column,** `Month Name = FORMAT ('Date Master Table'[Date], "mmm")`

**Add day column,** `Day = DAY ('Date Master Table'[Date])`

**Add Day name column,** `Day Name = FORMAT ('Date Master Table'[Date],"ddd")`

2. **Add Column Credit Type: add column (either add conditional column or write function or Dax)**

**Credit score:**

- Excellent: 800–850
- Very Good: 740–799
- Good: 670–739
- Fair: 580–669
- Poor: 300–579

```
Credit Score Status = SWITCH(TRUE(), Bank_Churn[CreditScore]>=800 && Bank_Churn[CreditScore]<= 850,
"Excellent",  Bank_Churn[CreditScore] >=740 && Bank_Churn[CreditScore] <= 799, "Very Good",
Bank_Churn[CreditScore] >=670 && Bank_Churn[CreditScore] <= 739 , "Good" ,  Bank_Churn[CreditScore] >= 580 &&
Bank_Churn[CreditScore] <= 669 , "Fair" ,  Bank_Churn[CreditScore] >= 300 && Bank_Churn[CreditScore]<= 579,
"Poor" )
```

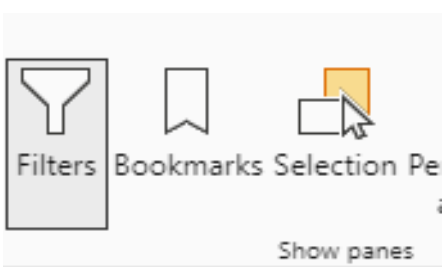
**Measures (DAX):**

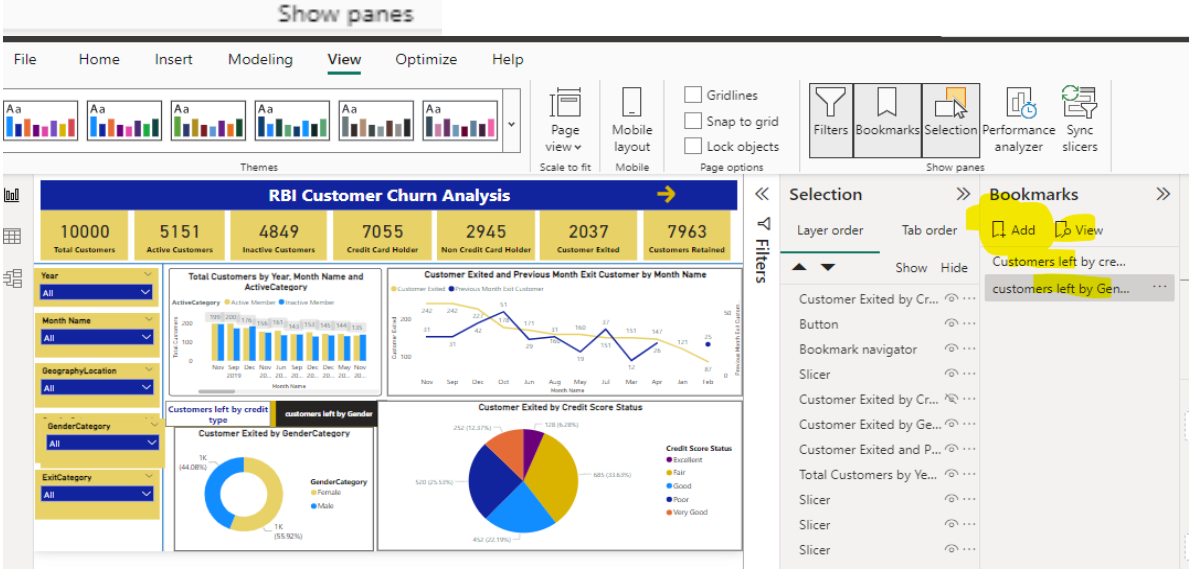
1. `Total Customers = count (Bank_Churn[CustomerId]) // This calculates total number of customers //`

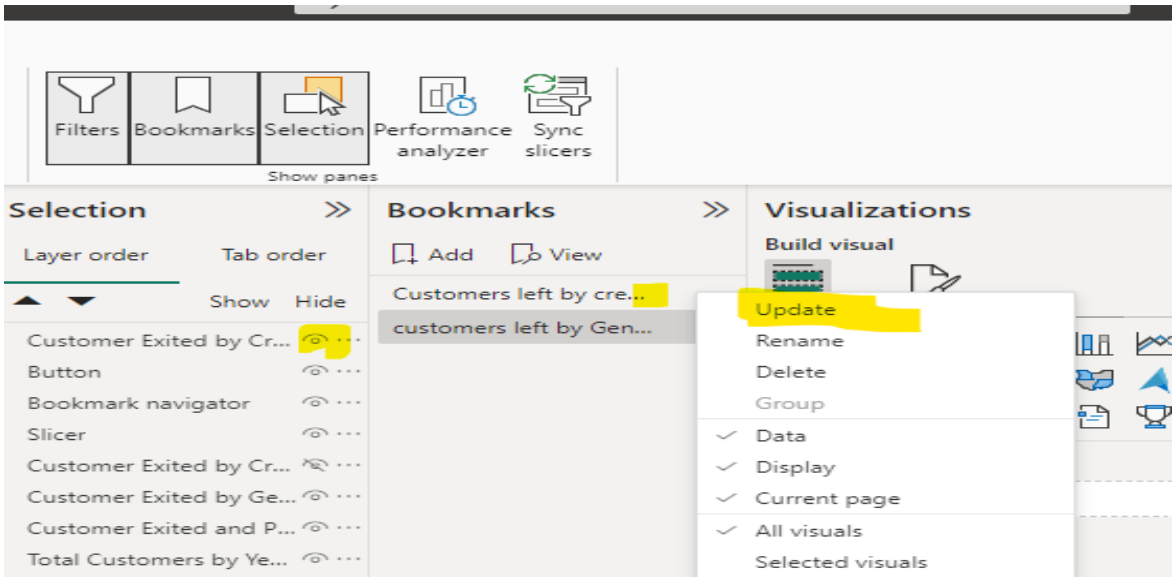
```
2. Active Customers = CALCULATE (count (Bank_Churn[CustomerId]), ActiveCustomer[ActiveCategory] =
   "Active Member")
3. Inactive Customers = [Total Customers]-[Active Customers] // this calculates total number of inactive
   customers in the organization//
4. Customer Exited = CALCULATE([Total Customers], 'Exit Customer'[ExitCategory] = "Exit") // this
   calculates no of customers exited from the total number of bank customers//
5. Customers Retained = CALCULATE([Total Customers], 'Exit Customer'[ExitCategory] = "Retain”) // this
   calculates total number of customers retained with the bank //
6. Credit Card Holder = CALCULATE (count(Bank_Churn[CustomerId]), 'Credit Card'[Category] = "credit card
   holder") // this calculates total number of customers having credit card from the bank//
7. Non Credit Card Holder = calculate (count (Bank_Churn [CustomerId]), 'Credit Card'[Category] = "non
   credit card holder") //this calculates number of employess doesn’t have credit card from the bank//
8. Previous Year Exit Customers = CALCULATE ([Customer Exited], PREVIOUSYEAR ('Date Master
   Table'[Date])) // this calculates previous year exited customers//
9. Previous Month Exit Customer = calculate ([Customer Exited], PREVIOUSMONTH ('Date Master
   Table'[Date])) // this calculates total customers exited previous month//
10. Previous Day exit customers = calculate ([Customer Exited], PREVIOUSDAY ('Date Master Table'[Date]))
   // this calculates previous day exit customers//
11. Churn % =
   var CE = [Customer Exited]
   var CT = [Total Customers]
   var Churnperc =DIVIDE (CE, CT)
   RETURN Churnperc    // we have added variable to write Dax//
```

- **Create a relationship (data modeling)**

- **Create a bookmark – Add Button >> or Navigator >> book mark navigator**
  When we use book mark? when we overlap tow charts or we want to hide same types of charts, we shuld add bookmarks to save dashboard space.



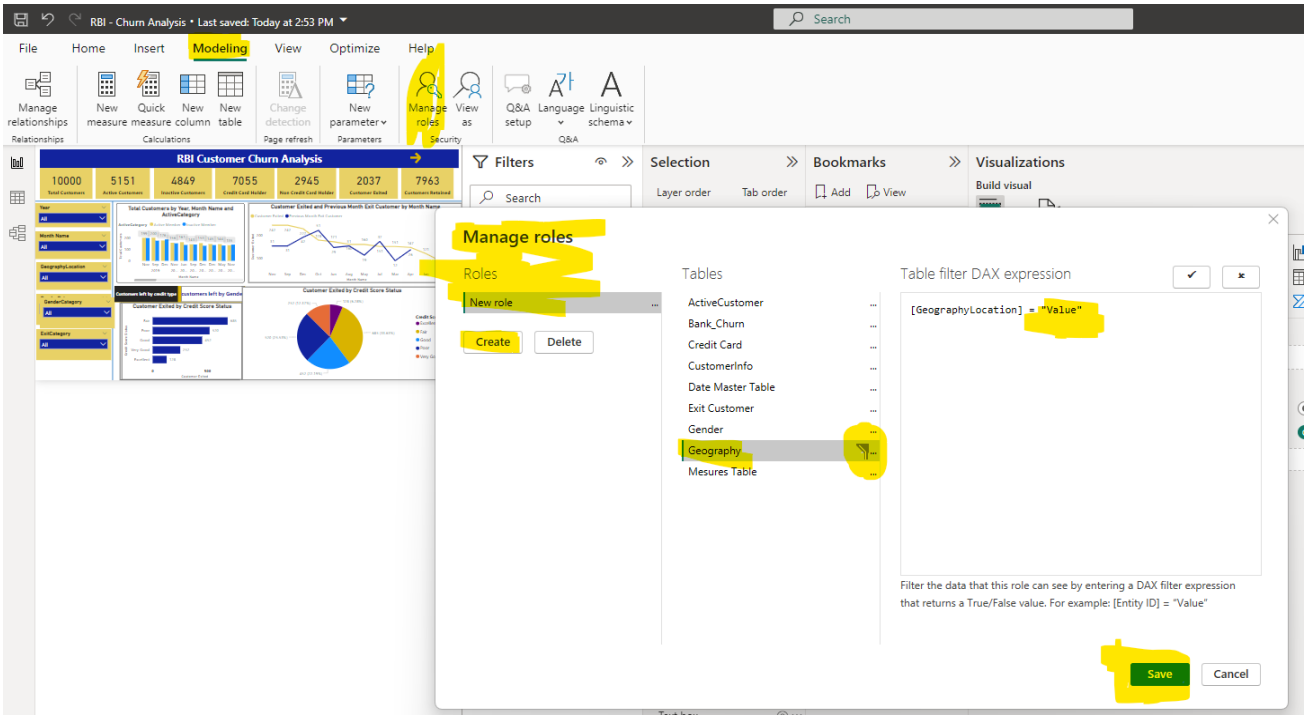



**Conclusion:**

Churn prevention allows companies to develop loyalty programs and retention campaigns to keep as many customers as possible.

## ***POWER BI SERVICES***

- **Row Level Security**
  This helps us to control data with users when we share the same file. Example: In case we have different county data we want an employees or customers to view only data of their respective country we must add row level security add country name under "value" (example - "Canada") save it will show Canada to the Canadian geography people.




- **Power BI Publish Report:** Power Bi Service >>>>Workspace >>>>> create workspace >>>>name of your project>>>>add email id >>>go to power bi desktop >>>publish

- **In your work space you can create a security & send an email to specific audience**

- **Data schedule refresh** >>> PB Service >>>>click on schedule refresh >>>>install gateway (if data is in your own system) no need gateway in case your data is in the cloud) >>>>select data file to connect to the gateways >>>>apply>>>> schedule refresh ON >>> set date & time >>>set refresh failure notification ON , add data set owner email id.
  **Note: 30 min interval is must to make data refresh, for PB pro data refresh allows 8 times in a day, for PB premium 48 refreshes allowed in a day with 30 min interval. If you need data to be refreshed less then 30 min or real time you must do the refresh manually with page refresh.**

- **Create dash boards in PB Service** – pin the dashboard visual, its not interactive, its static just an image. You can PIN what you want or just pin entire dashboard in desktop >>PB Service >>> create Dashboard >>> Pin what you want

- **PB Service** – **Subscription** – add email address & frequency

- **PB Service** – **Alerts** – manage alerts: alerts will help to get quick update about any changes in important data (in Bank churn analysis if I want to get an alert for number of Exited customers, I can set alert if any customer leaves the bank)
  Click on the card you want an alert for >>> Right Click to manage alerts >>>>add alert rule >>>> add alert parameter >>>> set email to be sent to the concerned person in the organization.

- **Application** – Mobile App (this is an end product) – File >>>> add title or name of project for mobile app (what name you will give to the app) >>>> it will be saved as DUPLICATE copy >>>>> Edit AS Mobile Layout

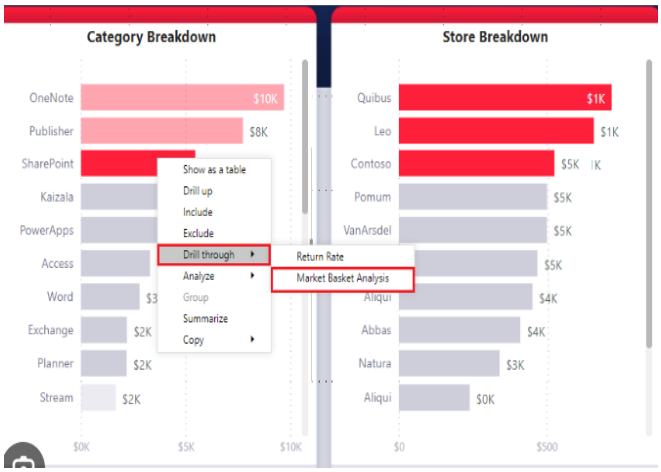
- **Imp steps to Create App ------ Set Up, Content, Audience.**
  Add content (show what you want to show to the audience) >>>>> Grant Access >>>>add email add specific users >>> Hide what you do not want to share with them in your application

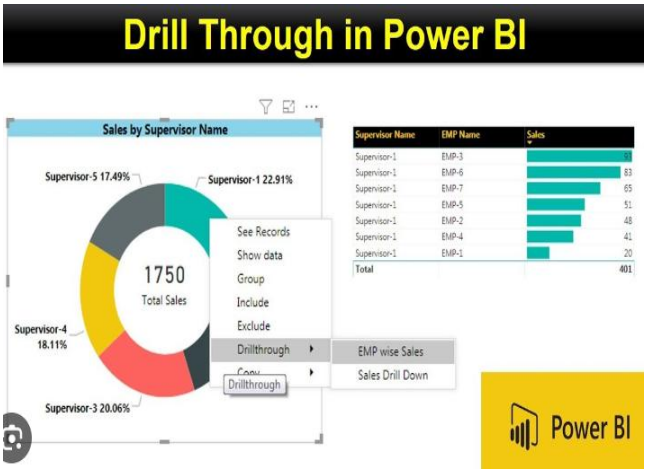
- **HOW TO ADD INTERACTIVE DASHBOARD IN YOUR PPT (**Power Point Presentation) ----
  Go to Power BI Pro >>>> Service >>>>Export >>>> PowerPoint >>>> Embed Live Data

- **Hierarchy:** right click on measures or columns or date >>> add hierarchy >>>Date, Year, Month, Quarter, Day

- **Drill Through:** You can create a drill through button in Power BI. This button drills through to a page with details that are filtered to a specific context. One way to drill through a report is to right-click in a visual. However, if you want the drill through action to be more obvious, you can create a drill through button instead

- What is the **difference between drill down and drill through** reports in Power BI?
  Drilling down can be achieved in a single view (i.e., you can drill down from the top level to the bottom level of the data in the same visualization). Drilling through uses a selected data item, may use filters, and applies them to a page you are drilling through.




- **Other Note: when you make changes to your report view dashboard, you need to publish again to see the changes in the final dashboard which you have shared with your client & other users. You must manually UPDATE the application from the power BI Service.**